Mathematical Induction

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Remember that the basic procedure of mathematical induction was as follows:

1. Prove that a statement S_1 is true.
2. Prove that, for all integers $n \geq 1$, that the statement S_n implies the statement S_{n+1}: $S_n \implies S_{n+1}$.
3. Conclude that S_n is true for all integers $n \geq 1$.

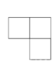
Use mathematical induction to prove the following statements:

1. Find and prove a formula for

$$\sum_{j=1}^{n} \frac{1}{j(j+1)}.$$

2. The sum of the first n squares of natural numbers is $\frac{n(n+1)(2n+1)}{6}$:

$$\sum_{k=1}^{n} k^2 = 1 + 4 + 9 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

3. Here is an L-shaped tromino:  . Show that, for all integers $n \geq 1$, a $2^n \times 2^n$ sheet of graph paper with one corner box removed can be tiled by such trominos.

Strong Mathematical Induction

Remember that the basic procedure of strong mathematical induction was as follows:

1. Prove that a statement S_1 is true.

2. Prove that, for all integers $n \geq 1$, that the statement S_k (for $k = 1, 2, \ldots, n$ implies the statement S_{n+1}: $\{S_1, S_2, \ldots, S_n\} \implies S_{n+1}$.

3. Conclude that S_n is true for all integers $n \geq 1$.

Prove the following using strong induction:

4. Recall that f_k, the Fibonacci numbers, are defined by $f_1 = f_2 = 1$ and $f_{n+1} = f_n + f_{n-1}$ for $n \geq 2$. Show that

$$f_n = \frac{\phi^n - \psi^n}{\sqrt{5}},$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the *golden ratio* and $\psi = \frac{1-\sqrt{5}}{2} = -1/\phi$.